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1 Appendix A: Sensilla Array Directionality and Beam Patterns

1.1 Objective

Quantify array directionality, beam patterns, and array gain for sensilla arrangements (log-periodic and uniform), relating morphology to directional detection.

1.2 Planned Methods (src)

- `src/antenna_arrays.py`
 - `compute_beam_pattern(wavelengths, positions, gains)`
 - `array_gain(pattern)`
 - `design_log_periodic_array(min_len_um, max_len_um, tau, count)`

1.3 Planned Script and Outputs

- Script: `scripts/generate_sensilla_array_directionality.py`
- Data: `output/data/sensilla_array.npz`
- Figure: `output/figures/sensilla_array_beam_patterns.png`
- Caption: `output/figures/sensilla_array_beam_patterns.caption.txt`

1.4 Figure

1.5 Equation References

- Effective aperture: see (??)
- Gain pattern: see (??)

1.6 Reproducibility

1. Run: `python3 scripts/generate_sensilla_array_directionality.py`
2. Artifacts saved to `output/data/` and `output/figures/`.

1.7 Cross-References

- Methods: ??

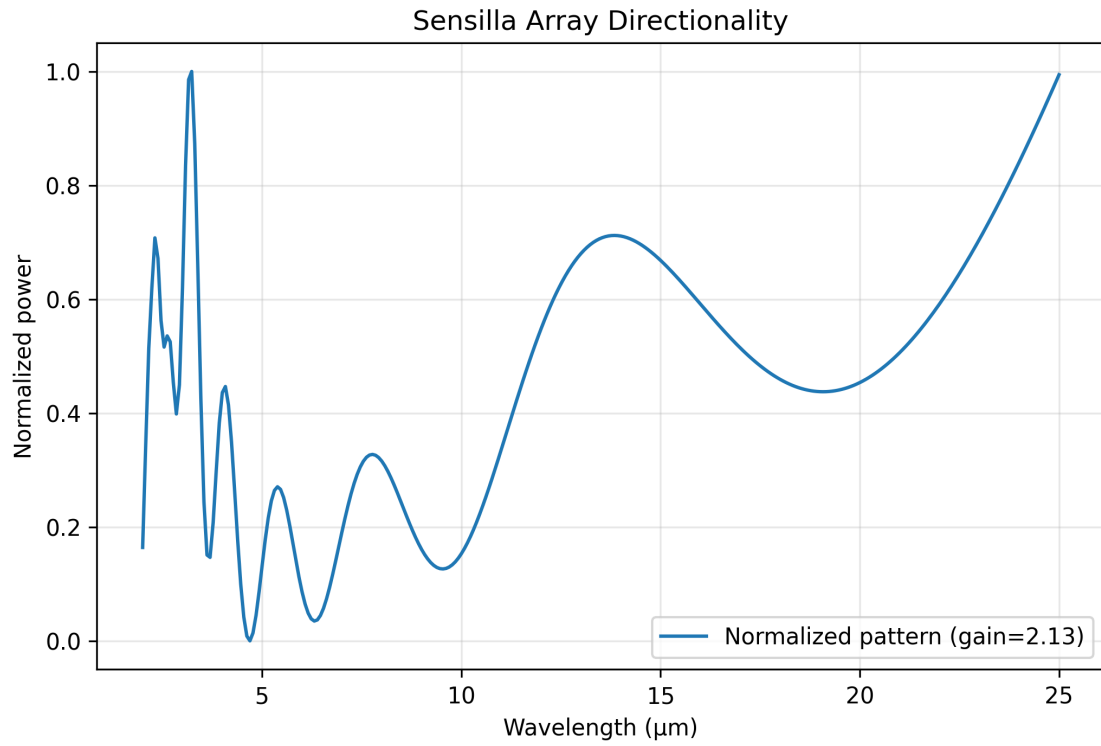


Figure 1: *Beam patterns for sensilla arrays across infrared wavelengths; log-periodic vs uniform arrays, showing half-power beamwidth and side-lobe structure.*

- Symbols: ??
- Mathematical forms: ??